

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2020
 Programme: BE Full Marks: 100
 Course: Microprocessors and Assembly Language Programming Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) How would you differentiate the microprocessor and microcontroller in term of frequency of operation, speed, size, complexity and structure? 7
 b) Draw block diagram of 8085 microprocessor. Explain each unit in brief. 8
2. a) What are the addressing modes of 8086 microprocessor? Explain with proper examples. 7
 b) Draw and explain a well-labelled timing diagram of the instruction STA5300H and calculate the time required to execute this instruction if the crystal frequency is 5 MHz. 8
3. a) Write an assembly language program in 8085 to calculate the factorial of the number up to 10 digits of a number given by user. 7
 b) There are two tables of the data stored at memory location A050H and A070H, each of which have 16 data, WAP to store data from first table to third table starting from A090H if corresponding data in first table is greater than second table else stored 00H in the third table 8
4. a) Explain the difference between Procedure and macro. Write an assembly language program to display a string 'POKHARA UNIVERSITY' characterwise using macro to insert space between characters. 8
 b) Design an address Decoding circuit to interface 4KB RAM, 8KB ROM and 16KB RAM with starting address 8000H. 7
5. a) What are the hardware and software interrupts of 8085 and 8086? Briefly describe the conditions which cause the 8086 to perform each of the following types of interrupts: Type 0, Type 1, Type2, Type 3 and Type 4. 8

- b) What is interrupt processing? Explain in detail about vectored and polled interrupts. 7
6. a) Explain the basic modes of operation of 8255. Determine the control word for the following configuration of Intel 8255: 8
 Port A- Input. Mode of Port A-Mode0
 Port B-Output. Mode of Port B-Mode0.
 Port C_{upper}-Output. Port C_{lower}-Input
 b) Write a subroutine program to generate a pulse of width 50 microseconds using 8254 PIT. Take frequency of clock as 2 MHz and use Counter 0. 7
7. Write short notes on: (Any two) 2×5
 a) Types of assemblers
 b) DMA controller
 c) Memory Classification